Peter Crew

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Education

- **University of Bath** — Bath, UK
  PhD in Statistical Applied Mathematics — *Oct 2024 - Present*
- **University of Warwick** — Coventry, UK
  MEng in Discrete Mathematics (First class) — *Oct 2020 - Jun 2024*
- **Nailsea School** — Nailsea, UK
  A-Levels - including Mathematics (A), Politics (A*), Physics (A)* — *Sep 2018 - Jun 2020*
- **Nailsea School** — Nailsea, UK
  *GCSEs - including Further Mathematics (A**), Mathematics (9), Computer Science (9)* — *Sep 2016 - Jun 2018*

Experience

- **SURE-AI** — Oslo, Norway
  Research intern — *September 2026 - December 2026*
  Working to develop an AI co-driven platform for mathematicians to share and explore results. Further work including optimal selection of models for agentic workflows.

- **University of Bath** — Bath, UK
  Teaching assistant — *October 2025 - Present*
  I run tutorials for 2nd year undergraduate students (analysis, algebra, numerical analysis)

- **AM Music** — Coventry, UK
  Piano teacher — *Jan 2024 - May 2025*
  One-on-one lessons with beginner students, focusing on the fundamentals of piano technique and reading music

- **MyTutor** — Online
  Mathematics Tutor — *Jun 2021 - Jun 2022*
  Responsible for preparing and carrying out lessons for GCSE pupils taking Mathematics

- **Self Employed** — Online
  Hired Programmer — *Sep 2017 - Sep 2019*
  Worked with customers to design and implement bot programs for the Discord application

Research Experience

- **Research Project** — University of Bath
  Inverse and implicit function theorems for rough differential equations — *Feb 2025 - July 2025*
  Under the supervision of Dr Avi Mayorcas, I explored the use of inverse and implicit function theorems in recovering the drivers of RDE solutions driven by geometric rough paths

- **Group Research Project** — University of Bath
  Spatial corrections for random forests in models of wildfire risk — *Feb 2025 - July 2025*
  As part of a group project modelling the growth of wildfires, I investigated autologistic corrections of random forest models used for predicting wildfire risk

- **Statistics Summer Internship** — University of Warwick
  The Inverse Problem for Rough Differential Equations — *Jun 2023 - Aug 2023*
  Supervised by Professor Anastasia Papavasiliou, I explored recent work of hers in estimating the noise from discrete observations of outputs of RDEs driven by rough paths

- **Dissertation** — University of Warwick
  Numerical Solutions of Stochastic Differential Equations — *Oct 2022 - May 2023*
  Supervised by Professor Aleksandar Mijatović, I wrote a survey of (strong/weak) numerical schemes approximating solutions to SDEs, along with some applications to options pricing and PDEs

Activities

- **SAMBa 2026 conference organiser**: I was one of four organisers for the 2026 edition of the annual SAMBa conference at Bath
- **Zero Gravity mentor**: Mentoring disadvantaged Sixth-Form students to improve their chances of gaining places at top universities
- **Treasurer of Warwick Real Ale Society 23/24**: Responsible for the finances of our annual tour and the largest student-run ale festival in Europe
- **GCSE Maths Volunteering**: During sixth form, volunteered to tutor struggling students taking Mathematics for GCSE
- **CREST Awards**: While in school, I received the Bronze, Silver and Gold CREST Awards on topics relating to interpretation of programming languages